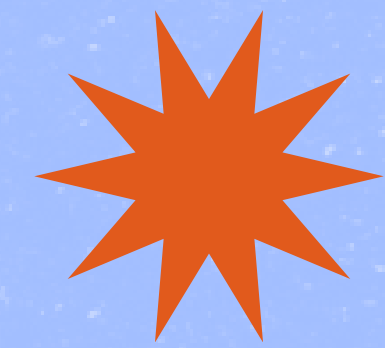
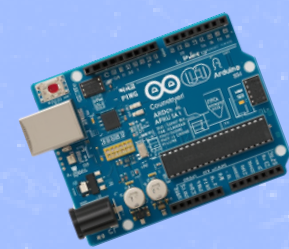
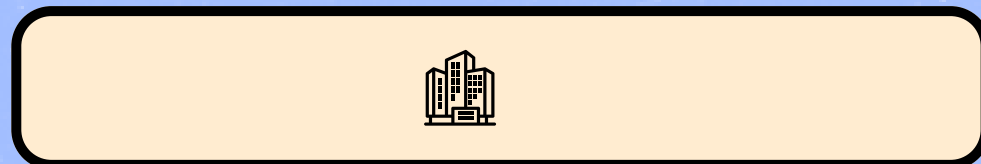
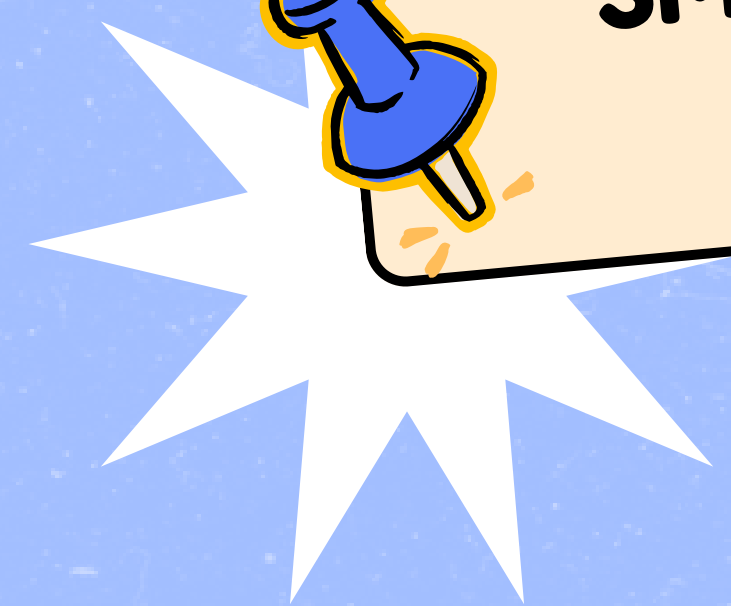




ARDUINO & BLUETOOTH (HC-05)

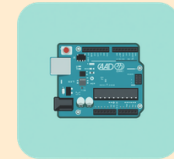


SMARTPHONE CONTROL





WHAT IS ARDUINO?

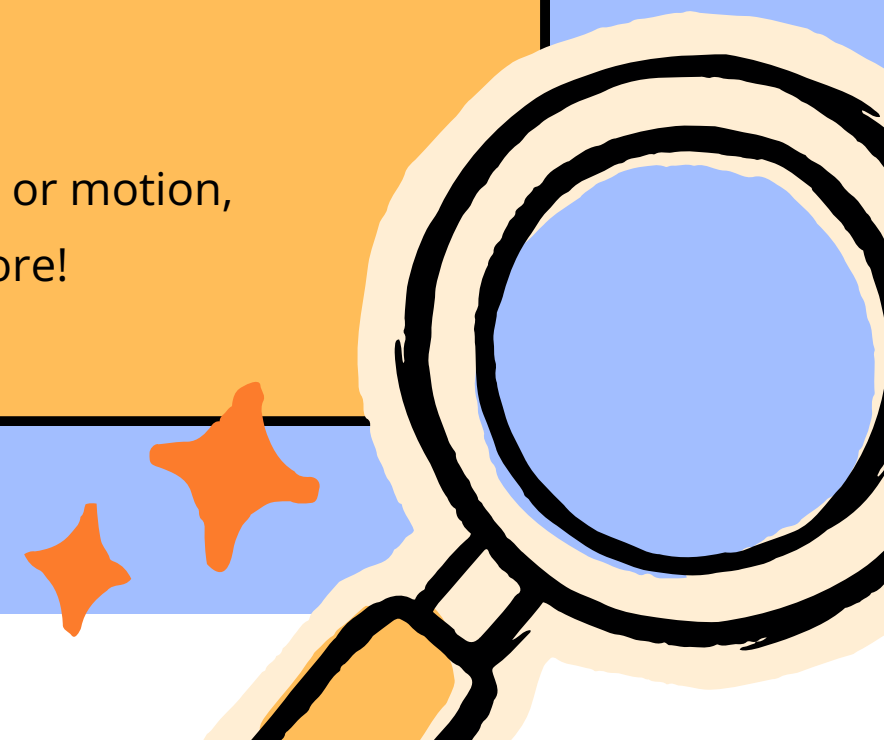


A Tiny Computer You Can Program!

Arduino is a small, friendly computer board that lets you control electronic things! You write simple code to tell it what to do, and it follows your instructions.

What Cool Things Can Arduino Do?

Control LED lights, spin motors, read sensors for temperature or motion, make robots move, create interactive games, and so much more!





WHAT IS BLUETOOTH?



Wireless Connection Magic!

Bluetooth lets devices talk to each other wirelessly when they're close by. We use it every day for headphones, speakers, and connecting phones!

Fun Fact: Named After a Viking King!

Bluetooth is named after Harald Bluetooth, a Viking king who united Denmark. Just like him, Bluetooth technology unites our devices!





MEET THE HC-05 BLUETOOTH MODULE



What is HC-05?

A small Bluetooth device that connects to your Arduino board. It lets your Arduino talk wirelessly to smartphones and other devices!

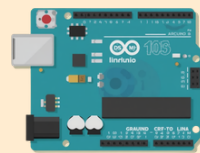
Key Specifications

Range: ~10 meters • Uses serial communication • Easy to connect • Has LED indicator, VCC, GND, TX, and RX pins





HOW DOES BLUETOOTH COMMUNICATION WORK?



Wireless Data Exchange

Arduino sends and receives data wirelessly through the HC-05 module. Your smartphone connects to HC-05, creating a two-way communication link!

Serial Communication Protocol

Data travels as serial signals between devices. Think of it like sending text messages—one character at a time, super fast!





CONNECTING HC-05 TO ARDUINO



Parts You Need

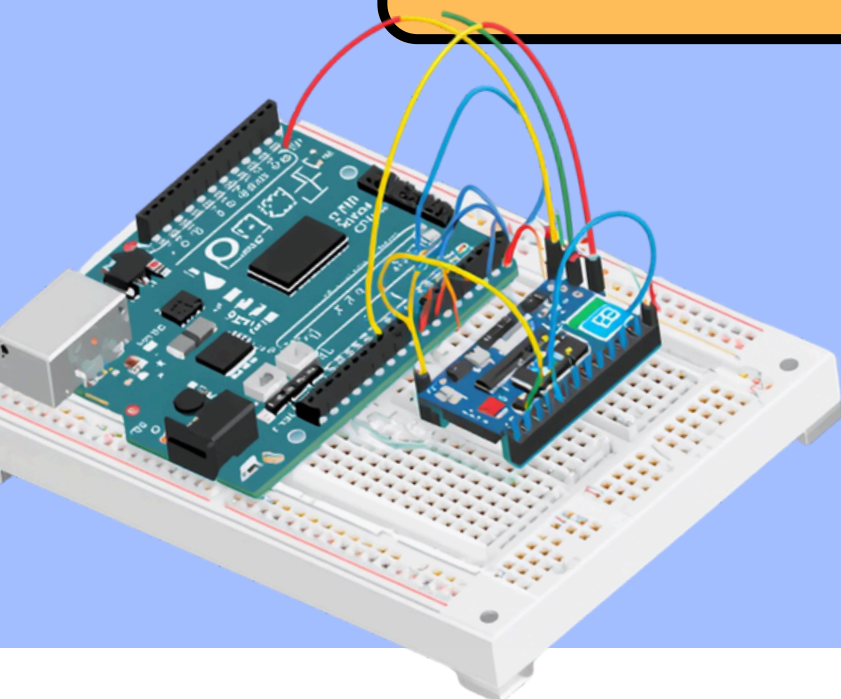
Arduino board, HC-05 module, jumper wires, and a breadboard

Wiring Steps

Connect VCC to 5V, GND to GND, TX to RX, and RX to TX pins

Safety First!

Always power off your Arduino before connecting any wires!





PAIRING HC-05 WITH YOUR SMARTPHONE

**Step 1: Turn on HC-05
and Phone Bluetooth**

**Step 2: Search and
Select "HC-05"**

**Step 3: Enter Password
"1234"**

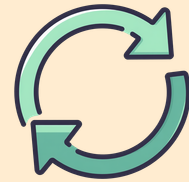
**Wait for LED to Blink
Slowly**

**You're Now
Connected!**





WRITING BASIC ARDUINO BLUETOOTH CODE



Setting Up Serial Communication

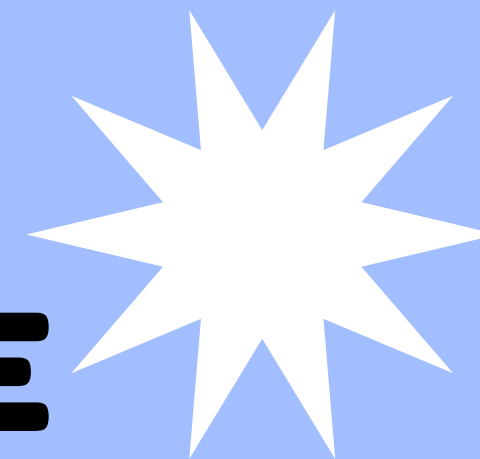
In `setup()`, use `Serial.begin(9600)` to start communication. This tells Arduino to talk at 9600 speed—like setting the right channel on a walkie-talkie!

Understanding the Main Loop

In `loop()`, use `Serial.read()` to listen for phone commands. When data arrives, Arduino reads it and can turn things on or off based on what you send!



APPS TO CONTROL ARDUINO VIA SMARTPHONE



Bluetooth Electronics

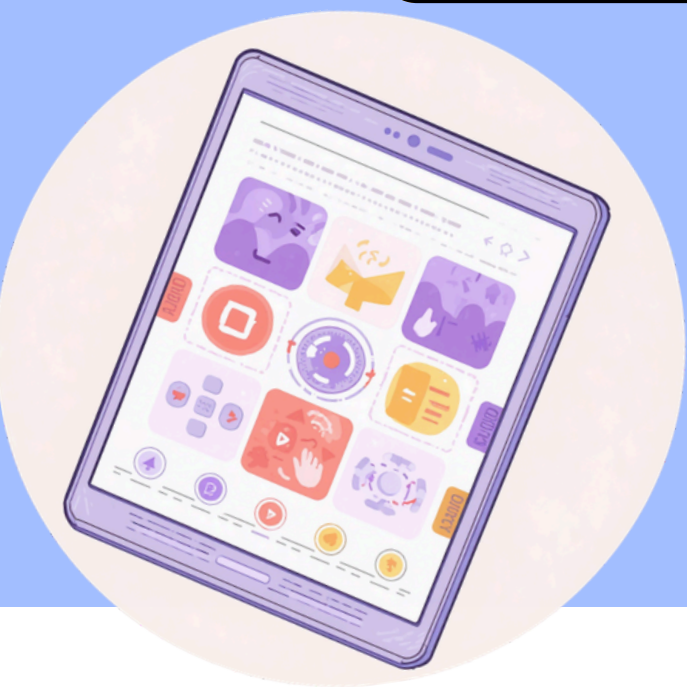
Create custom buttons, sliders, and displays to control your Arduino projects

Arduino BT Controller

Simple interface with ready-made buttons to send commands to Arduino

Easy Bluetooth App

Beginner-friendly app with touch controls for LED and motor projects





PROJECT 1: CONTROL AN LED WITH YOUR PHONE

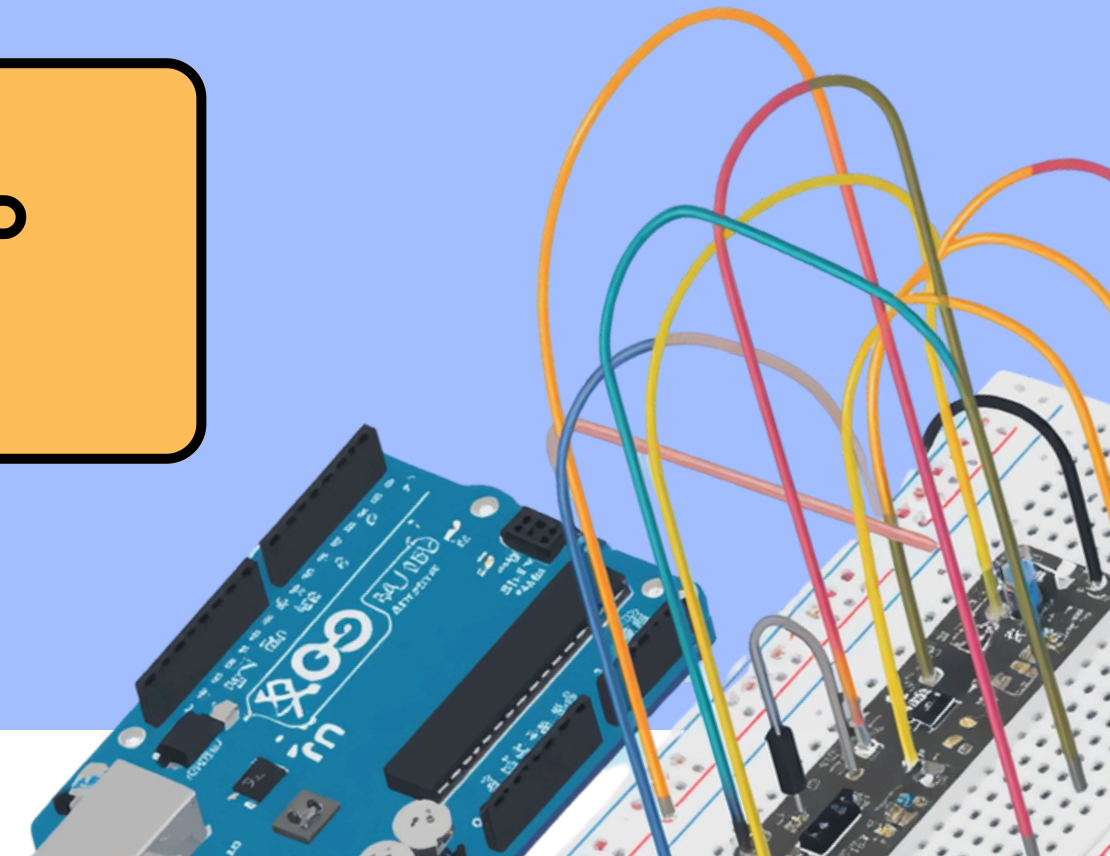
Parts Needed: Arduino, HC-05, LED, Resistor, Wires

Wiring: Connect HC-05 TX to Pin 0, RX to Pin 1

Code: Use Serial.read() to check for '1' or '0'

Send '1' from phone app to turn LED ON

Send '0' to turn LED OFF!





PROJECT 2: MOTOR CONTROL

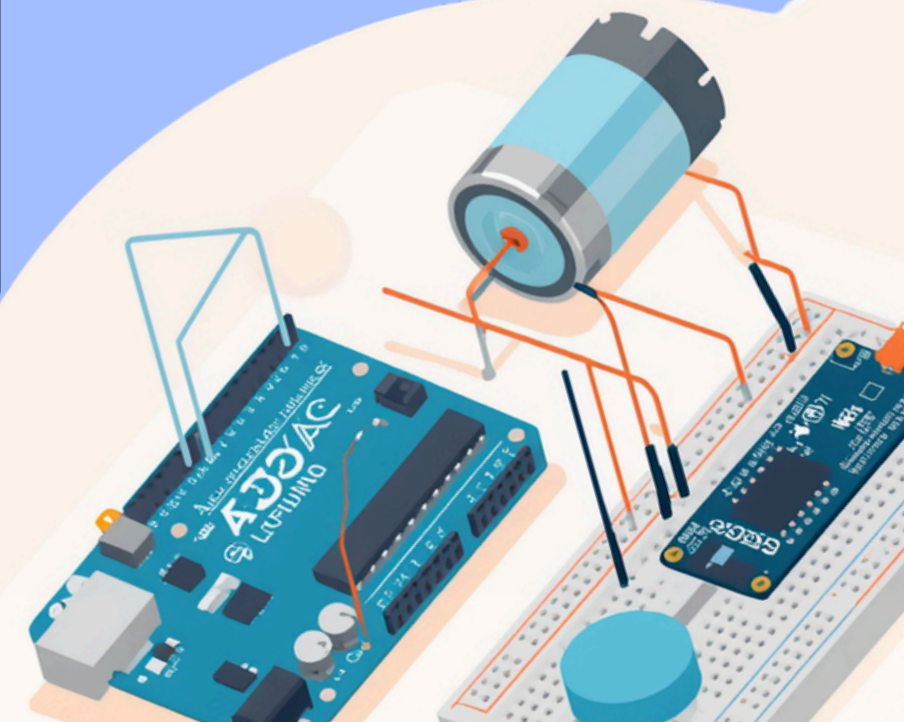
Parts Needed: Arduino, HC-05, DC Motor, L298N Driver

Wiring: Connect motor to driver, driver to Arduino pins

Code Controls Speed and Direction via Bluetooth

Send 'F' for Forward, 'B' for Backward, 'S' to Stop

Watch Your Motor Spin Wirelessly!





SAFETY AND BEST PRACTICES



Disconnect Power

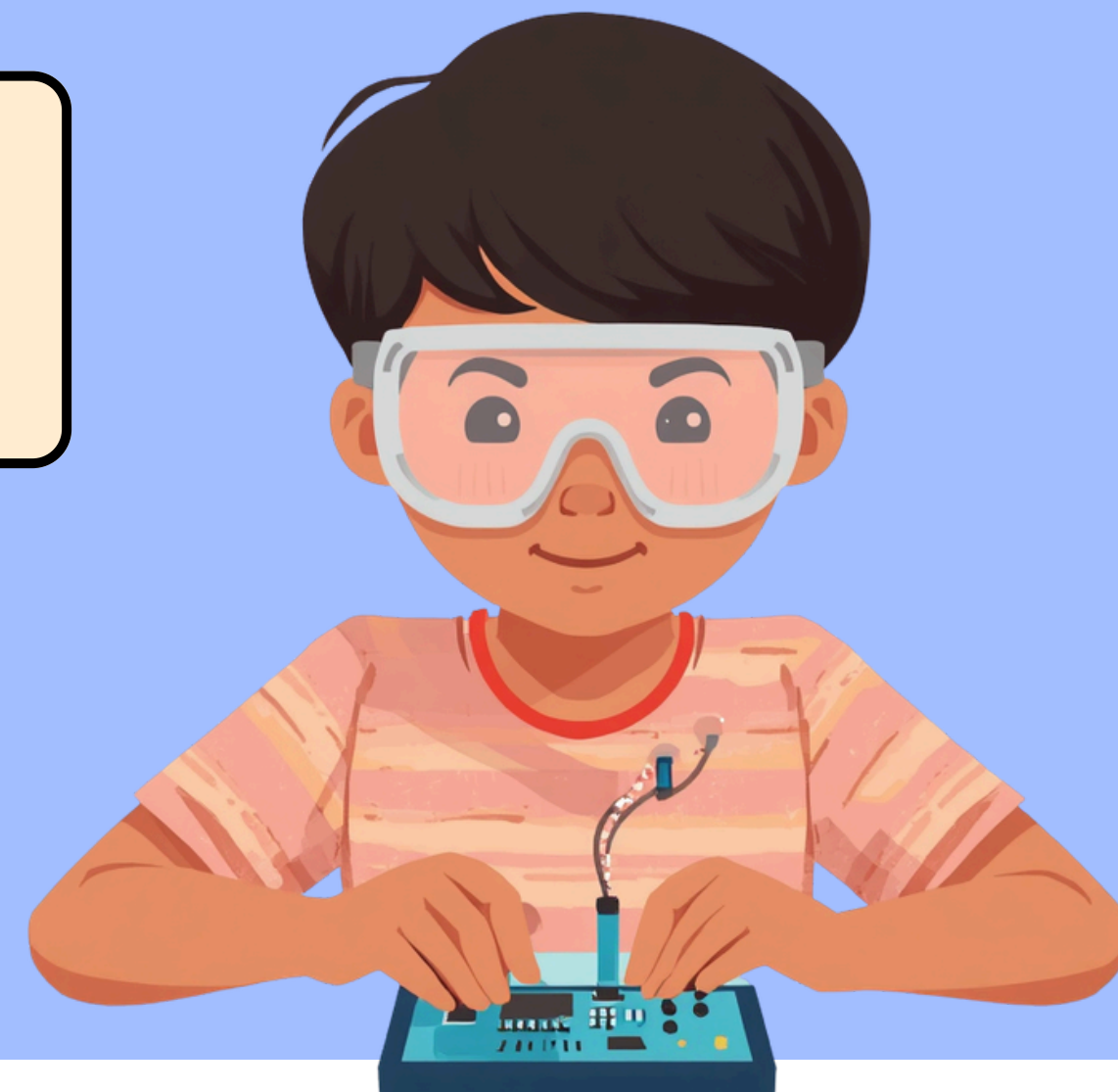
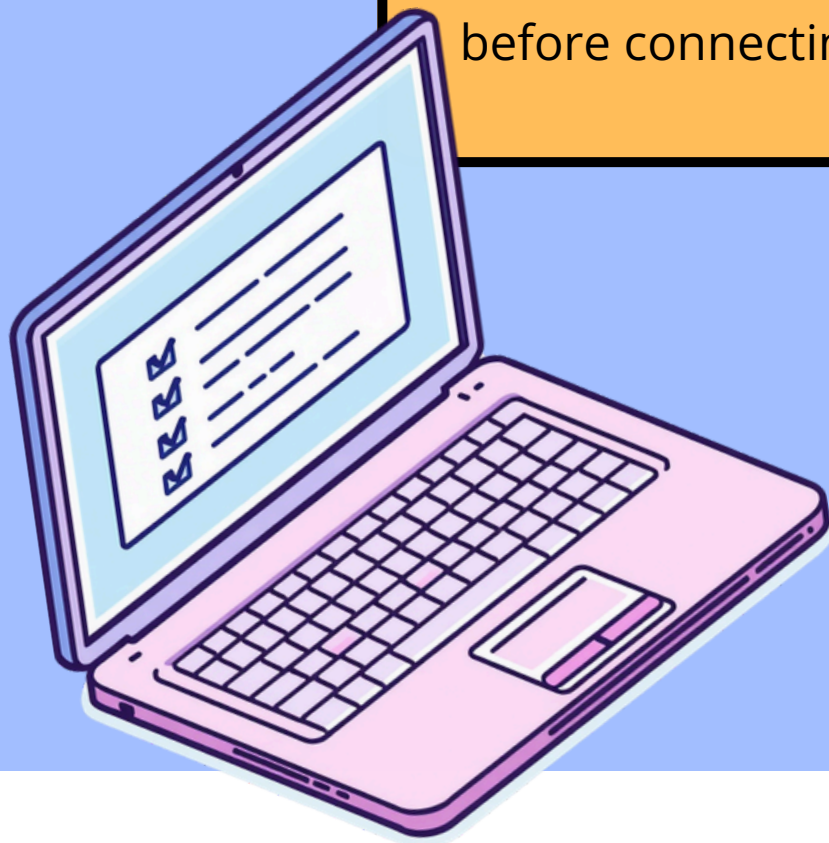
Always turn off and unplug your Arduino before connecting or changing any wires

Protect Components

Use resistors with LEDs and handle batteries carefully to avoid damage

Keep Water Away

Never let water or liquids near your electronics projects





TROUBLESHOOTING COMMON PROBLEMS

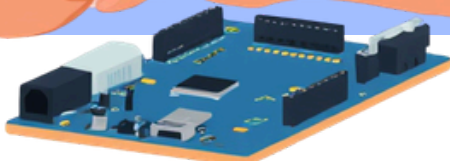
HC-05 Not Pairing? Check power and Bluetooth status LED

Arduino Not Responding? Verify wiring and code upload

LED/Motor Not Working? Check all connections and power

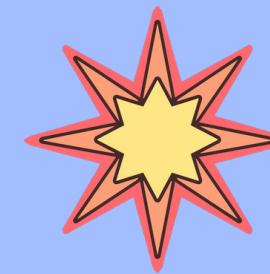
Reset HC-05 Module Hold button while powering on

Don't Give Up! Ask for help!





FUN BLUETOOTH PROJECT IDEAS!



Remote-Controlled Car

Build a car you control from your phone!

Bluetooth Robot Arm

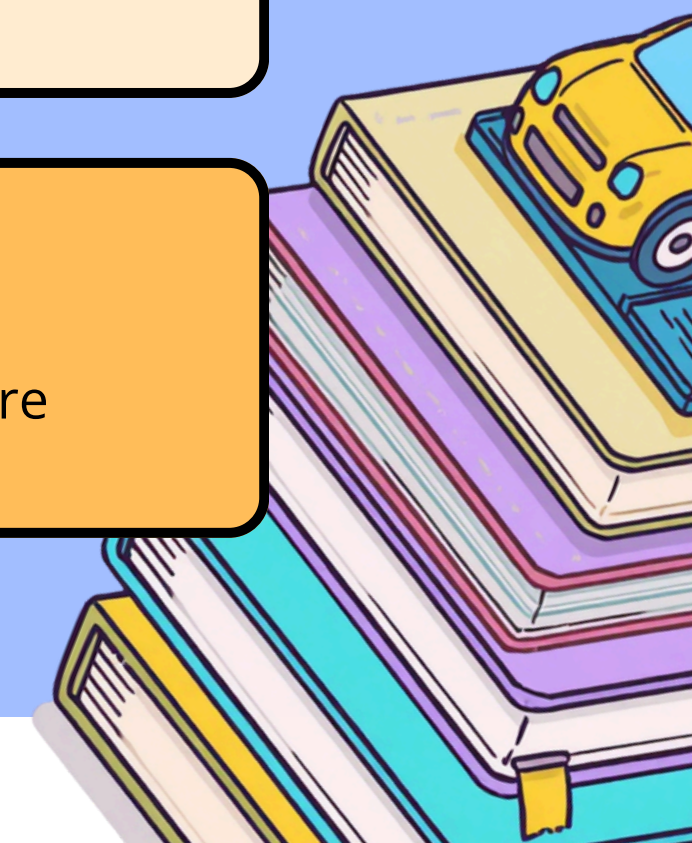
Move objects with phone commands

Wireless Home Lighting

Turn lights on and off wirelessly

Smart Fan Control

Adjust fan speed from anywhere





SUMMARY & WHAT YOU LEARNED

Arduino & HC-05 Basics
Arduino is a mini computer that controls things wirelessly via HC-05 Bluetooth!

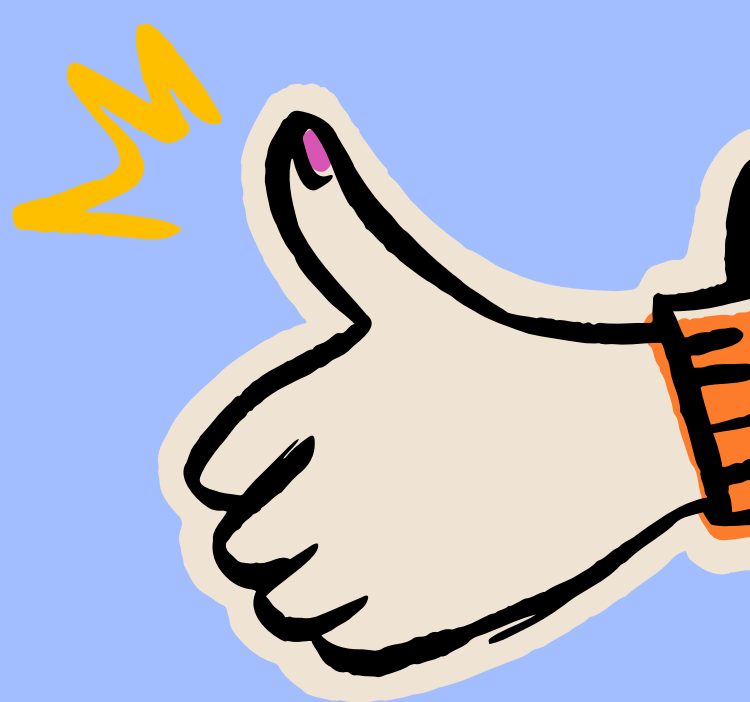
Smartphone Connection
Pair HC-05, use apps to send commands, and watch your projects respond!

Projects You Can Build
LED control, motor control, robots, smart home gadgets and more!

Stay Safe & Experiment
Disconnect power when wiring, use resistors, and keep exploring!

Keep Building Amazing Things!





QUESTIONS & LET'S BUILD TOGETHER!

Ready to create your own cool Bluetooth projects? Ask your questions, share your ideas, and let's start building amazing wireless creations with Arduino and HC-05!

